

MA 513: Ordinary Differential Equations

Course Code	MA 513
Course Name	Ordinary Differential Equations
L-T-P-C	3-1-0-4
Prerequisites	NA
Intended for	M.Sc. /M.S./Ph.D./B.Tech 3rd and 4th year
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **General Overview** [8 Lectures]
Solutions methods: General solution methods, Power Series methods with properties of Bessel functions and Legendre polynomials.
- **Existence and Uniqueness** [11 Lectures]
Existence and Uniqueness of Initial Value Problems: Picard's and Peano's Theorems, Gronwall's inequality, continuation of solutions and maximal interval of existence, continuous dependence
- **Systems of Differential Equations** [16 Lectures]
Algebraic properties of solutions of linear systems, The eigenvalue-eigenvector method of finding solutions, Complex eigenvalues, Equal eigenvalues, Fundamental matrix solutions, Wronskian, Matrix exponential, Nonhomogeneous equations, Variation of parameters, Stability theory for linear and nonlinear systems, Lyapunov function.
- **Boundary value problems** [7 Lectures]
Green's function, Sturm comparison theorems and oscillations, eigenvalue problems.

Text Books

1. G.F. Simmons and S.G. Krantz, **Differential Equations: Theory, technique and practice**, Tata McGraw-Hill, 2007.
2. V. Arnold, **Ordinary Differential Equations**, MIT Press, 1978.
3. Coddington, E. A. and Levinson, N., **Theory of Ordinary Differential Equations**, Krieger Publishing Co, 1984.

References

1. Ahmad, S. Rao, M.R.M., **Theory of ordinary differential equations with applications in biology and engineering**, EWP publication, 1999.
2. L. Perko, **Differential Equations and Dynamical Systems**, *Texts in Applied Mathematics*, Vol. 7, 2nd Edition, Springer Verlag, 1998.
3. Devaney, R., Hirsch, M. W. and Smale, S., **Differential Equations, Dynamical Systems, and an Introduction to Chaos**, 2nd Edition, Academic Press, 2003.
4. Birkhoff, G. and Rota, G.-C., **Ordinary Differential Equations**, Wiley, 1989
5. R.P. Agarwal and D. O'Regan, **An Introduction to Ordinary Differential Equations**, Springer- Verlag, 2008.